DOUBLE JEOPARDY — The Vulnerability Spiral

Compound Climate Risk Across Five Island Nations

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The Question

Small island nations face a compounding climate vulnerability — high physical exposure to sea-level rise, layered with degrading natural coastal defenses (mangroves and coral reefs). Climate adaptation funding is often allocated using single-indicator exposure metrics alone. Is that reliable, or does it systematically misrepresent true risk? And do all ecosystem buffers degrade uniformly, or does that assumption itself need testing?

The Method

Five island nations across three ocean basins — Canary Islands, Fiji, Lakshadweep, Maldives, Seychelles — were tested using 10+ independently-sourced datasets spanning 1996–2024. Physical exposure (settlement-level sea-level-rise risk) and coral thermal-stress trend (NOAA Coral Reef Watch Degree Heating Week) were normalized and combined into a single Compound Vulnerability Score. Mangrove extent was tracked separately across three independent time points (1996, 2010, 2020) using Global Mangrove Watch — treated as a distinct hypothesis, not assumed to move with coral reefs.

The Finding

Physical exposure alone is misleading. The Maldives has the highest sea-level-rise exposure of any island tested (99.1% of settlements at risk) — yet Seychelles emerges as the highest overall-risk island once ecosystem degradation is factored in, driven by the most severe coral thermal-stress trend recorded across the sample.

Hypothesis	Result
H1 — Coral reefs are degrading	Supported — significant for Maldives ($p=0.011$) & Seychelles ($p=0.025$)
H2 — Mangroves are degrading	Not supported — zero net decline across 3 islands, 3 time points
H3 — Governance tracks vulnerability	Suggestive only — $r=0.727$, $p=0.164$ (not significant)
Compound Vulnerability Score (top)	Seychelles 0.880 · Maldives 0.651 · Lakshadweep 0.467

The Compound Vulnerability ranking is stable across a full 0–100% weighting-sensitivity sweep — Seychelles stays highest-ranked up to ~74.5% physical-exposure weighting, well past the 50/50 weighting actually used.

Validation & Robustness Checklist

- ✓ Two ecosystems tested independently — no uniform-decline assumption
- ✓ Mann-Kendall trend test on the full 24-year coral series (not just period comparison)
- ✓ 3 independent time points for mangroves (1996 / 2010 / 2020)
- ✓ Population-weighted exposure recomputation as a cross-check
- ✓ Full 0-100% weighting-sensitivity sweep (compound score robustness)
- ✓ Honest null result reported — H2 (mangrove decline) not supported
- ! Governance correlation flagged as not statistically significant (suggestive only)

Honest Limitation

The governance-alignment test (protected-area coverage vs. verified vulnerability) shows a moderately strong positive correlation ($r=0.727$) but does not reach statistical significance at this sample size ($p=0.164$, 95% CI: -0.43 to 0.98) — a genuine limitation of testing only five islands, reported as suggestive rather than confirmed. The Compound Vulnerability Score also uses min-max normalization, meaning scores are relative within this five-island sample rather than absolute — Canary Islands' score of 0.000 reflects the lowest raw values in this sample, not the absence of risk.

Real-World Relevance

Climate adaptation funding for small island nations is frequently allocated using single-indicator exposure metrics. This project's central methodological argument — that exposure and true compound risk are not the same thing, and that ecosystem buffers do not degrade uniformly — has direct implications for how conservation and adaptation resources should be prioritized across islands and ecosystem types.

GitHub: github.com/sakshimaske303-commits/DOUBLE-JEOPARDY | **Live Dashboard:** double-jeopardy-6ev9trz3dwafsb7panbnxg.streamlit.app | **Zenodo DOI:** 10.5281/zenodo.21739961

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